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INSTITUTO SUPERIOR TÉCNICO
Departamento de Engenharia Informática
Forensics Cyber Security
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Digital Forensics Report Lab2

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1 Acquired artifacts

Name	Type	SHA-256 Value
johnny_bash_history.txt	SHA-256	C368617EF9982AE9BE3F776584AC1DAE533E83B9460DBC10DA5C14B460153324
gitconfig.txt	SHA-256	345FDB845A85735F71775DD873D17DC6C36A98FA2B07325124BFFDE8EE71A983
thebasement.09-23.log	SHA-256	0ED38C48752362A3747EBF027B86864B2950E427B1688CC343D9257682538945
auth.log	SHA-256	47c8b7c3132827f578b68ed07fc90e447a3d1a4f70bad6b40945dc1e06f3e52a
seed.txt	SHA-256	98b073309a566415455c7f57b56b6d280fde814147dae61363363632e5a5fc47

2 Report of all findings

Initially, we used the command `$ mmls backup.img` to see which partition might contain the content relevant to our investigation concerning this server backup.

DOS Partition Table					
Offset Sector: 0					
Units are in 512-byte sectors					
	Slot	Start	End	Length	Description
000:	Meta	0000000000	0000000000	0000000001	Primary Table (#0)
001:	-----	0000000000	0000002047	0000002048	Unallocated
002:	000:000	0000002048	0039684095	0039682048	Linux (0x83)
003:	-----	0039684096	0039686143	0000002048	Unallocated
004:	Meta	0039686142	0041940991	0002254850	Win95 Extended (0x0f)
005:	Meta	0039686142	0039686142	0000000001	Extended Table (#1)
006:	001:000	0039686144	0041940991	0002254848	Linux Swap / Solaris x86 (0x82)
007:	-----	0041940992	0041943039	0000002048	Unallocated

We began our investigation of this backup by examining **node 2048** using the command `fls -o 2048 backup.img`. After analyzing and reviewing the results, we found a directory **ironcaesar** containing several ZIP files with server code backups. These files were extracted and later confirmed to be backups of the **ZYRA website**.

```
(joaolucas@kalijlucas)-[~/Transferências]
$ fls -o 2048 backup.img 130562
d/d 130906: johnnymusk
d/d 130567: ironcaesar

(joaolucas@kalijlucas)-[~/Transferências]
$ fls -o 2048 backup.img 130906
r/r 130908: .bash_logout
r/r 130909: .bashrc
r/r 130910: .profile

(joaolucas@kalijlucas)-[~/Transferências]
$ fls -o 2048 backup.img 130567
r/r 130912: .bash_logout
r/r 130913: .bashrc
r/r 130914: .profile
d/d 130915: backups
r/r 130916: .bash_history
d/d 130917: .ssh

(joaolucas@kalijlucas)-[~/Transferências]
$ fls -o 2048 backup.img 130915
r/r 130919: backup_1758495226.zip
r/r 130920: backup_1758585623.zip
r/r 130921: backup_1758586202.zip
r/r 130922: backup_1758586801.zip
r/r 130923: backup_1758587401.zip
r/r 130924: backup_1758588001.zip
r/r 130925: backup_1758588601.zip
r/r 130926: backup_1758589201.zip
```

Once inside Johnny's workstation file system, we immediately noticed the **.bash_history** file, which stores the record of commands executed by the user. This file contains commands showing the use of VSCode for editing, copying of sensitive files, execution of tools such as irssi, use of the secure deletion tool srm, and deployment commands. It serves as the primary source for reconstructing the sequence of actions performed including the manipulation of files previously obtained during the initial forensic examination of the website.

```

1 |cd ~/Documents/ZYRA/
2 |ls -l
3 |code .
4 |irssi
5 |ls -l /media/johhnymusk/Intenso/
6 |ls -l
7 |ls -l resources/
8 |cd resources/
9 |cp /media/johhnymusk/Intenso/* .
10 |cat diff.txt
11 |xdg-open FinancialReport.pdf
12 |xdg-open Invoice.pdf
13 |file parking.db
14 |code parking.db
15 |xdg-open TouristThreads.png
16 |xdg-open wpp.png
17 |irssi
18 |git clone https://github.com/shadowexploitthunder-web/HideFiles.git
19 |ls -l HideFiles/
20 |code .
21 |chmod +x HideFiles/create_wordlist.sh
22 |chmod +x HideFiles/create_zip.sh
23 |chmod +x HideFiles/hide_base64.sh
24 |chmod +x HideFiles/hide_in_comments.sh
25 |chmod +x HideFiles/pdf_to_png.sh
26 |ls -l ~/Music/
27 |python3 HideFiles/wav_hider.py -h parking.db ~/Music/MingleGame.wav MingleGame.wav
28 |rm parking.db
29 |rm ~/Music/MingleGame.wav
30 |ls -l ~/Pictures
31 |HideFiles/hide_in_comments.sh diff.txt ~/Pictures/BarragemFagilde.tiff ~/Pictures/BarragemOdelouca.jpg
32 |rm diff.txt
33 |rm ~/Pictures/BarragemFagilde.tiff
34 |rm ~/Pictures/BarragemOdelouca.jpg
35 |ls -l ~/scripts
36 |source ~/.venv/bin/activate
37 |HideFiles/hide_base64.sh Invoice.pdf ~/scripts/steghide steghide
38 |rm Invoice.pdf
39 |rm ~/scripts/steghide
40 |cp ~/Pictures/fst-gang.png .
41 |python3 HideFiles/lsb.py -m hide -d horizontal -c rgb -n 3 -o fst-gang.png -p wpp.png -e png

```

```

42 rm wpp.png
43 rm ~/Pictures/fst-gang.png
44 HideFiles/pdf_to_png.sh FinancialReport.pdf blank
45 rm FinancialReport.pdf
46 cp ~/Documents/Modern_Cars.pdf .
47 HideFiles/create_wordlist.sh Modern_Cars.pdf
48 rm ~/Documents/Modern_Cars.pdf
49 HideFiles/create_zip.sh wordlist.txt fst-gang.png ~/Pictures/222.jpg ~/Pictures/456.jpg blank
50 rm ~/Pictures/222.jpg
51 rm ~/Pictures/456.jpg
52 rm blank
53 rm fst-gang.png
54 rm wordlist.txt
55 ls -l
56 cp ~/Pictures/RaceCar.png .
57 python3 HideFiles/lsb-complex.pyc -m hide -d diagonalup -c rgb -n 4 -o RaceCar.png -p TouristThreads.png
58 rm TouristThreads.png
59 rm -rf HideFiles/
60 rm ~/Pictures/RaceCar.png
61 rm .gitkeep
62 deactivate
63 keepassxc &
64 code ~/Documents/ZYRA
65 ls -l
66 git add --all
67 git commit -m "finally finished"
68 git push
69 cd ~/scripts/
70 chmod +x deploy.sh
71 ./deploy.sh
72 irssi
73 cd ~/Documents/
74 sudo apt install secure-delete
75 srm -r *
76 cd ..
77 ls -l
78 srm zyra.zip

```

The .gitconfig file also contained the same credentials that had been identified in the previously analyzed repository.

Inside the virtual environment `/.venv` we identified several Python packages that can be abused by malicious actors. The packages **keyboard** (1458646) and **sshkeyboard** (1458777) were found: **keyboard** can act as a keylogger, capturing everything typed, while **sshkeyboard** allows remote keyboard control over SSH and can be used to interactively control a compromised system. We also detected **pynput** (1459615), which provides advanced keyboard and mouse control and monitoring APIs and is often used in more sophisticated malware to automate input or record user actions. The **evdev** package (1459234) was present as well; it gives low-level access to input devices (keyboards, mice) and can intercept system input events outside normal user-space channels. We found **cryptography** (1459673), a library of strong cryptographic primitives legitimate for secure development, but also potentially used to encrypt stolen data or implement obfuscation/ransomware. For web communication, **requests** (1459370) and **urllib3** (1458721) were present; these libraries facilitate transmitting data over the internet and can be used for data exfiltration or contacting command-and-control servers. Finally, several PDF-processing tools (**pypdfium2**, **pdfminer**, **pdfplumber**) were identified; they enable advanced document processing and extraction of

information from PDFs, which can be useful for harvesting sensitive data contained in confidential documents.

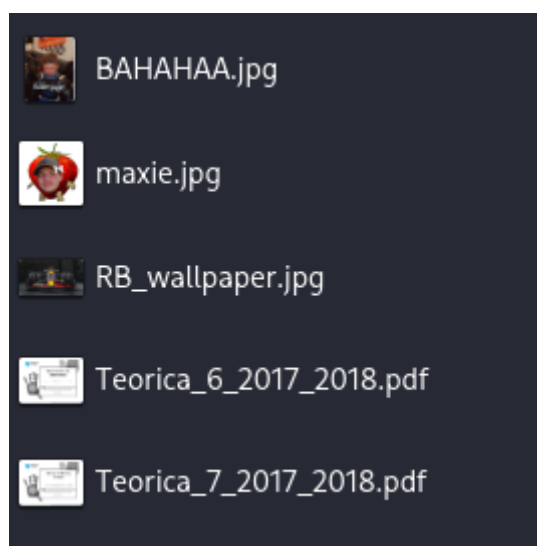
Inside `/.ssh` we also identified a `config` file linking GitHub identities notably **MuskyBoi** and **ironcaesar** along with associated SSH keys. These keys correspond to the server responsible for hosting the ZYRA website code.

We also discovered `scripts_hacking.zip`, `wordlist_gen.py`, and multiple `HideFiles/*.sh` scripts. Context: Tooling cloned from a repository (references to a repo such as “shadowexploit” / `HideFiles`). The collection includes `wordlist_gen.py` used to generate password lists and several shell scripts that automate file hiding, obfuscation, and packaging operations.

```
└─$ fls -o 4096 johnny.img 1443157
r/r 1443328: cookieSteal.html
r/r 1443329: deploy.sh
r/r 1443330: exploit.html
r/r 1443331: exploit.py
r/r 1443332: reflectedXSS.html
r/r 1443333: source.py
r/r 1443335: wordlistFromPdf.py
```

Inside the `irssi` directory we located `thebasement.09-23.log`. This IRC log records conversations between **johnnymusk** and **CatDucky**, including admissions of intent, operational instructions, and references to a repository used for hiding files. The log therefore serves as corroborating evidence of coordination and collaboration between the participants.

Inside the `Desktop` folder we also found the car image used for steganography, as well as course materials from IST for the Forensic Cybersecurity class covering steganography, watermarking, and storage/memory analysis.



Inside the `backups` folder we located four files of critical importance that reveal how the ZIP archives and their passwords for the ZYRA website backups, which we previously recovered from the server backup, were generated and packaged.

We also searched within the `tmp` folder for the file `seed.txt`, which is likewise used in the generation of the ZIP file passwords.

We returned to the directory where we had placed the server backup ZIPs and created three scripts. First, we decompiled the obfuscator using `decompyle3` in a `pyenv` environment running Python `3.8`, which produced `obfuscator_decompiled.py`. Using the algorithm recovered from that file, we developed `correct_password_generator.py` to brute-force the ZIP passwords by iterating the seed through multiple rounds. Finally, we wrote a shell script (`test_passwords.sh`) to test the generated passwords against the archives.

```
import hashlib, sys
SEED_PATH = "/tmp/seed.txt"
with open(SEED_PATH, "r") as f:
    line = f.readline()
if len(line.split("\t")) == 2:
    n = int(line.split("\t")[0])
    seed = line.split("\t")[1].strip()
    if n == 0:
        print("For the first run, please just place your password in the " + SEED_PATH + " file.")
        exit(1)
    else:
        n = 0
        seed = line.strip()
pw = hashlib.sha256(str(seed + str(sys.argv[1])).encode("utf-8"))
print(pw.hexdigest())
next_seed = hashlib.sha256(str(seed).encode("utf-8"))
with open(SEED_PATH, "w") as f:
    f.write(str(n + 1) + "\t" + next_seed.hexdigest())
```

```

def regenerate_seed_history(initial_password, iterations):
    """Regenera o histórico do seed desde o início"""
    current_seed = initial_password
    for i in range(iterations):
        current_seed = hashlib.sha256(current_seed.encode()).hexdigest()
    return current_seed

def generate_zip_password(timestamp, seed):
    """Gera a password para um ZIP específico"""
    # Algoritmo: SHA256(str(seed + timestamp))
    data = str(seed + str(timestamp)).encode('utf-8')
    return hashlib.sha256(data).hexdigest()

# Configuração
INITIAL_PASSWORD = "TheByteOf78"

backups = [
    ("1758495226", 69),
    ("1758585623", 70),
    ("1758586202", 71),
    ("1758586801", 72),
    ("1758587401", 73),
    ("1758588001", 74),
    ("1758588601", 75),
    ("1758589201", 76),
]

print("=== TESTANDO DIFERENTES ITERAÇÕES ===")

for timestamp, base_iteration in backups:
    print(f"\nTimestamp: {timestamp}")

    # Testar iterações próximas
    for offset in [-3, -2, -1, 0, 1, 2, 3]:
        test_iteration = base_iteration + offset
        if test_iteration >= 0:
            seed_used = regenerate_seed_history(INITIAL_PASSWORD, test_iteration)
            password = generate_zip_password(timestamp, seed_used)

            print(f"  Iteração {test_iteration}: {password}")

```

The final investigation report documents a coordinated scheme of criminal activity centered around the company MEMEL, marketed as a “smart parking” service. Our analysis shows that the company’s pricing algorithm was deliberately manipulated. The scheme operated to systematically overcharge tourists, to favor accounts associated with collaborators, and to create artificial “premium” parking zones that unjustifiably increased fees for ordinary users. This manipulation constitutes organized algorithmic fraud.

In parallel, the investigation uncovered evidence of money-laundering activity linked to a shell company named Futuro Inteligente, Lda. Although presented externally as an AI consultancy, this entity functioned as a conduit to move and obscure illicit proceeds effectively serving as a laundering channel for the funds derived from the parking scheme.

The case also involves conspiracy and corruption among multiple participants who coordinated via encrypted email services. Communications indicate the use of privacy-focused providers such as TutaMail and ProtonMail to exchange sensitive information and to conceal operational details from oversight. These encrypted exchanges corroborate a pattern of deliberate secrecy and coordination among the actors.

3 Analysis of relevant findings

3.1 Do the seized disk images contain traces of the phishing website (source code, deployment scripts, logs) and/or the hidden artifacts discovered in Lab 1? If so, identify them, their locations on disk, and their SHA-256 values as recovered from the images.

Yes, upon careful search inside the image “johnny.img”, a bash command log, or history (“.bash_history”), was uncovered in the directory “home/johnny”. This consisted in a text file with a significant amount of bash shell commands that are possibly related to the findings of Lab1.

First of all, there is mention of the name “ZYRA” in the files directories present in the log. Furthermore, the command “code” is issued several times with the current working directory as the target, which indicates usage of the VSCode IDE, possibly to code harmful scripts. Apart from that, the file structure of the website is present in this bash log, since many of the same directories and files found in the website have been mentioned, structured as expected, according to the observed directory structure in the website. Examples include the resources directory, which was crucial to the first Lab investigation, and, inside this one, the files previously downloaded and extracted from the website, namely the parking database “parking.db” (SHA-256: 23108D6552BBD633ED80C3DE50449E0698294C0DA3A049A61BBAED41E0EDCEA0), the whatsapp png file “wpp.png” (SHA-256: C938A05B84F2D749FC6AC65A88F57CC0C860D4169577947425CC01AA1BBD1C3), previously extracted from within another password-protected png file (“fst-gang.png”, also present in this log), and other such compromising files:

- FinancialReport.pdf (SHA-256: FDE6E8582B5B4763E4E574C5EB293876F5894339215D12BFFD224F929E559CC9)
- Invoice.pdf (SHA-256: 748E7358CCBBA0B769B94ECB3C96C161FD8366454ACF076A6F95B5F404515E1D)
- TouristThreads.png (SHA-256: FF56D339F5AFFE0279530214DEC0EF01950828BEB4AF6BFA83FE8465A71DA7D8)
- MingleGame.wav (SHA-256: 61D0C9689692F25CC21DD1F52F1A127120DDFF9D830A81853F6A24417DB0492A)
- BarragemOdelouca.jpg (SHA-256: A67C91879DC59568F098C13D03339FB4D5B6D6CFCE4CAEDA7C5340692A639BEA)
- BarragemFagilde.tiff (SHA-256: E171F9250DE647F16917EAE5A888DDC614AA470A10D1C261B77A4D768F2F286C)
- steghide (obfuscated pdf masked as binary) (SHA-256: ECCD76A28840E5CDF54A301060E5B5CB3BEC89C6F5105A8F7CC9F3BD5ABC3E63)
- Modern_Cars.pdf (SHA-256: F536596C90572AD983571C5432E52F2A4E3DE5784EF1923BC9D2FDD82033C9FB)
- myzip.zip (SHA-256: 35C5CA580DF0B730E044CD2705FD135731E30D00A6E8F9DB8439819E2BC0158D)
- RaceCar.png (SHA-256: D4DEB9F64E94DDC34BC65552F39B417462C7A437C30D826B415CEF843A241AAD)
- diff.txt (SHA-256: 80373BC7994731717F3FE753685848541BB81CBF906A85C3F42415E5ED05BD86).

All of these files were recovered from inside a directory named “/home/johnnymusk” either in the “johnny.img” or “backup.img” images.

This indicates the intent to hide information from forensics experts (see 3.3). Moreover, there are various bash shell scripts also mentioned within these logs, labeled in such a way that likely denotes their intended use. If this is the case, these were surely the scripts used to hide, alter, manipulate and deploy the information uncovered in Lab1 (see 3.2).

Additionally, the recovered workspace contains a complete phishing toolchain: client-side exfiltration pages (*ReflectedXSS.html*, *CookieSteal.html*) and a template escape (*exploit.html*) that explicitly target *document.cookie* to harvest session tokens, deployment automation (*deploy.sh*) that packages and publishes the site, and server-side exploitation tools (*exploit.py*, *source.py*) capable of executing commands and exfiltrating sensitive files. The co-location of these artifacts in `/home/johnnymusk/` establishes a direct operational link between development activity on the seized workstation and the phishing/credential-theft infrastructure. Furthermore, when considering the explicit statements of intent by the user “johnnymusk” ([See here](#) and [here](#)) regarding phishing activities, together with the presence of these specific tools on the seized disk, the cumulative evidence strongly supports the conclusion that the user was actively involved in initiating and carrying out the detected phishing operations.

Finally, files containing credentials (both email and password) ([See here](#)) were found multiple times inside zip files contained within the “backup.img”, in the directory (for each of the unzipped backup folders) `/home/johnnymusk/Documents/ZYRA/`. These zip folders were located in the directory `ironcaesar/backups/` within the mentioned image. This is a major indicator of potential phishing activities.

For these reasons, it is believed these are undeniable traces of the phishing website, directly connecting the apprehended workstation image to the malicious VM, thus strengthening the case against João Musk.

3.2 Can you reconstruct the provenance and handling of these files? Provide a timeline of relevant events that shows how the phishing site was built and operated, and how the hidden artifacts were managed on the seized machines.

It becomes reasonably straightforward to piece together the setup of the phishing website, and previously uncovered secret files, by following the commands present in the “bash_history” file. In the beginning, the user enters the directory `~/Documents/ZYRA`, proceeding to use the “code” command, calling an IDE (VSCode), and then issuing the command “irssi”, which corresponds to an internet relay chat client program. This indicates both the intentions to code and also to communicate via a more secure, decentralized and private channel. Eventually, the user enters the directory `resources/` and proceeds to copy the contents of `media/johnnymusk/Intenso/` to the current working directory (`resources/`). The commands that follow immediately shed light on the nature of the contents copied, and these are revealed to be the secrets uncovered in the Lab1 investigation, such as “parking.db”, “Invoice.pdf”, among others already mentioned in Section 3.1.. Later, a git repository is cloned into the working directory. This git repository is stated to have useful tools to aid in the activity of file hiding. This information is present in a text conversation, made possible using the tool “irssi”, and can be found in the file “thebasement.09-23.log”, found within the directory `home/johnnymusk/` ([See here](#)). It is worth mentioning that the frequency of occurrence of the command “irssi” in the bash log is very coherent with the text conversation log “thebasement.09-23.log”, due to the nature of the information discussed in the text conversation and files and commands tested/known up until each command in the log. The narrative adds up across these two pieces of evidence.

The contents of the folder “HideFiles” correspond to bash scripts that are then made executable by the user, and utilized to perform information concealing and steganography, directly matching what had been observed in the artifacts analysed in Lab1. It is possible to observe all instances of data obfuscation, steganography and information concealing previously described in the Lab1 report. Namely:

- The parking.db file being hidden within the MingleGame.wav file;
- The image files ("BarragemOdelouca.jpg" and "BarragemFagilde.tiff") being injected with the contents of "diff.txt";
- The peculiar steghide binary being injected with the base_64 encoded contents of the "Invoice.pdf" file;
- The WhatsApp conversation image "wpp.png" being injected, via LSB (Least Significant Bit) Steganography, within the "fst-gang.png" file;
- The "FinancialReport.pdf" file being obfuscated and tampered with to resemble a png and trick file identification software;
- The "Modern_Cars.pdf" file being used to generate a wordlist that would be used to provide a password for the newly created zip file containing not only the "fst-gang.png" file, but also two other JPEG files, "222.jpg" and "456.jpg", that are strongly believed to have been placed alongside real evidence in order to confuse forensic investigators, and the "blank" file, previously "FinancialReport.pdf";
- The "TouristThreads.png" file being injected via LSB Steganography inside the file "RaceCar.png" (using the EXACT distribution pattern employed previously for extraction in the occasion of the investigation pertaining to Lab1).

The above mentioned observations erase any possibility of a "coincidence", regarding the provenance and handling of the files critical to this investigation. These point unequivocally to the authorship of "João Musk" behind the presence of the secret artifacts in the ZYRA clone website.

All of the compromising files and tools used were removed by the user and the remaining contents were subsequently pushed to a remote repository.

Lastly, the "deploy.sh" script ([See here](#)) was made executable and run. This is the script responsible for copying the contents of the "~/zyra.zip" file to the remote directory "csf:/home/muskyboi" (using the command "scp") and consequently launching the website.

The contents of the "/Documents/" folder were then erased by using the command "srn", as well as the "zyra.zip" file that had all other clues. The usage of the command "srn" is particularly concerning, as it permanently deletes files and directories by overwriting their contents multiple times before removing them, thus rendering the data virtually unrecoverable. This alludes to a dire need of employing extreme anti-forensic strategies, most likely due to the sensitive nature of the data being manipulated.

3.3 Did you find any evidence of anti-forensic activity? If so, describe the techniques observed and the evidence that supports your claim.

Yes, we found some traces that indicate clear attempts of anti-forensic activities within the sized disk images (johnny.img and backup.img).

File obfuscation and steganography were present in the files "fst-gang.png" and "wpp.png" and they both were previously identified as containing hidden data.

The ".bash_history" file from the directory /home/johnny references tools and commands consistent with data hiding operations.

The file "fst-gang.png" contained embedded data later identified as part of a greater scheme, demonstrating deliberate obfuscation.

The use of misleading and encrypted filenames such as "blank" and "steghide" appear to have contents inconsistent with their extensions suggesting the rename or repackaging to avoid detection. The presence of such files points directly to anti forensic manipulation.

In the “backup.img”, under the directory /home/ironcaesar/, commands such as mkdir backups and exit were used and followed by a nearly empty or truncated .bash_history. This suggests manual cleanup of shell history to reduce the possibility of incrimination.

In the file “johnny.img” multiple commands were present and that indicates a possible deletion or use of shell history suppression, for example unset HISTFILE or editing .bash_history to avoid forensic recoveries.

The directory /home/ironcaesar/backups/ contains copies of files from other users, probably restructured to disguise their original context, and this could be interpreted as an attempt to stage false trace making it difficult to know the true source of the phishing operation.

3.4 What new discoveries emerge from the seized images that were not visible in Lab 1? Do these findings strengthen the attribution of the phishing attack to João Musk, or point to other relevant actors (collaborators, external servers)? Based on your analysis, what hypotheses can you form about the larger scheme hinted at by the hidden artifacts, and what recommendations would you make for further investigative steps by Prof. Alexandre Magno and the Polícia Judiciária?

The new discoveries not present/visible in Lab1 are:

- The cloned git repository used to access file hiding tools;
- The “deploy.sh” script, detailing how the cloned website went live, and other compromising scripts;
- The “srm” commands used to erase data permanently;
- The backups present in “home/ironcaesar/backups/”, which are the exact artifacts retrieved from the website;
- The conversations between the users johnnymusk and CatDucky, both by using the “irssi” tool and by e-mail ([See here](#));

All of the critical evidence above was found on João Musk’s personal computer (the johnnymusk account). The connection between these artifacts and the phishing operation is clear. Crucially, the irssi logs contain an admission by johnnymusk stating an intent to “make ZYRA suffer” by stealing users’ credentials. Taken together, the technical evidence and the explicit confession strongly indicate that João Musk is the principal actor behind the phishing scheme.

Evidence tying João Musk to the operation:

- Critical artifacts (scripts, thebasement.09-23.log, .bash_history, and the ZYRA directories) were recovered from /home/johnnymusk.
- The chronological sequence in .bash_history links the build, obfuscation, packaging, deployment and cleanup steps to the johnnymusk account.
- Hashes of incriminating files (e.g., fst-gang.png, wpp.png, parking.db) appear only in locations associated with johnnymusk activities and the commands used by that account.

Evidence pointing to third parties:

- IRC logs and the deploy.sh script reference a remote host named muskyboi and an associated user; these entries indicate a remote entity received the deployment package.
- Complete copies of the same artifacts in /home/ironcaesar/backups/ suggest at least one other person had access to or possession of the same material.
- The git clone mentioned in the irssi log points to an external repository (not local) that appears to host the toolkit and workflows used; this suggests a channel for sharing tools or coordination with others (e.g., a private server or hosted git service).

The new findings substantially strengthen the attribution to João Musk: the combination of local artifacts, operational commands, communication logs and an explicit admission create a robust linkage. Nevertheless, the presence of remote hosts and backups stored under other accounts strongly indicates that João Musk likely did not act entirely alone; there is probable cooperation with at least one other person and/or use of third-party infrastructure.

The recovered evidence suggests that the phishing operation is likely part of a larger and more coordinated scheme involving multiple actors and distributed infrastructure. While João Musk appears to be the technical originator and local operator, several findings indicate that other individuals or servers were also involved, fulfilling complementary roles.

- Examination of `.bash_history` and the `deploy.sh` script shows that João Musk locally obfuscated and packaged the website as `zyra.zip` within the directory `/home/johnnymusk/Documents/ZYRA/`. The script then executes an `scp` command to upload this archive to a remote host named `muskiboi` (e.g., `scp zyra.zip csf:/home/muskyboi/`) and remotely unpacks it to launch the phishing website. This workflow clearly divides responsibilities: João handled content preparation and obfuscation on his own workstation, while the remote server hosted the live, malicious website. Such separation is a common operational security measure, since it limits local exposure and shifts forensic evidence and liability to the external server.
- The cloned git repository found in `/home/johnnymusk/Documents/ZYRA/` contained a structured set of scripts (`HideFiles/*.sh`, `deploy.sh`, `wordlist_gen.py`, etc.) that automate file obfuscation and steganography. The presence of documentation and standardized naming conventions strongly implies that this was not an ad-hoc creation by João Musk alone. Instead, it points to a shared toolkit, likely exchanged among members of a small technical group or even sold in underground communities. The conversation logs in `thebasement.09-23.log` reinforce this idea the user `CatDucky` refers to “the repo with the hiding tools” ([See here](#)) and discusses updates to scripts, implying shared maintenance and collaboration.
- The repeated use of steganography, secure deletion (`srm`), and shell history manipulation indicates a mature anti-forensic strategy. These methods not only obscure evidence on the local system but also facilitate the replication of the scheme on other infrastructures. The combination of data obfuscation, encrypted zips, and distributed backups (e.g., `/home/ironcaesar/backups/`) suggests an intent to replicate and redeploy the phishing framework across multiple environments while maintaining deniability.
- It is plausible that João Musk acted primarily as the developer and packager of the phishing content, while other entities such as `muskiboi` or `ironcaesar` were responsible for hosting, distribution, or backup operations.

The forensic structure supports this hypothesis:

- `johnnymusk's` account shows intense development activity (editing, encoding, obfuscation, deployment).
- `ironcaesar's` account mainly contains replicated copies of final artifacts in `/home/ironcaesar/backups/`, with little or no evidence of code creation.
- The remote host `muskiboi` (mentioned in multiple deployment commands) acted as the execution node that published the live phishing site.

Together, these roles form a small, compartmentalized cell designed for operational resilience and plausible deniability.

On another note, regarding the larger scheme hinted at by the hidden artifacts, new information has been uncovered. Previously, the remaining doubts regarding the scheme were essentially tied to whether MEMEL had knowledge of the operation, what was the nature of “Futuro Inteligente Lda.”, and what contribution could be attributed to the members identified in the WhatsApp group screenshot. Some light has been shed on this matter, by uncovering a hidden e-mail conversation ([See here](#)), that is strongly believed to pertain to the same actors present in the “irssi” conversation, present below as well ([See here](#)).

This e-mail was sent by the account “itscatducky@proton.me”, using Proton Mail, designed for privacy. This hints at the potentially compromising nature of the contents. The destination is user “sussyjohnnyhacker@gmail.com”. The e-mail essentially involves a forwarded e-mail exchange between sender “totallyaveragedude@tutamail.com” and receiver “itscatducky@proton.me”. In this e-mail, a scheme involving MEMEL and “Futuro Inteligente Lda.” is apparently exposed. The e-mail states that “Futuro Inteligente Lda.”, the primary beneficiary of MEMEL’s financing, is merely a ghost company used for the activity of money laundering. The e-mail appears certain that the AI Research and Development activities supposedly carried out by the company are a façade to “clean” the money generated by the discriminatory practices in parking pricing, described in the first report. The exchange also mentions a name, “Trocós”, that is, according to the information conveyed in that exchange, the receiver of MEMEL’s “investment”. The e-mail suggests the actors involved in this e-mail exchange are not affiliated to MEMEL or enablers of the scheme. Further investigation is needed.

This has serious implications. Assuming, for a moment, that the information is to be trusted, this would clear many doubts and suspicions. Firstly, it would confirm MEMEL’s knowledge, and even endorsement, of the operation. Moreover, it would unveil the true purpose of “Futuro Inteligente Lda.”, meaning that this company would not be stealing from MEMEL at all, rather being used to enable the malicious practices instead. This exchange also points towards and reinforces the necessity of interrogating the actors mentioned in the previous report, pertaining to the WhatsApp group screenshot. Furthermore, the previously uncovered secret artifacts become easily explainable under these claims. Not only is the “blank.pdf” file (first report’s appended artifacts - MEMEL Financial Report) not stolen from MEMEL, it is, in fact, the motivation needed to justify the actions claimed to have been carried out by MEMEL itself. The Invoice (first report’s “Receipt.pdf” appended artifact) would now likely constitute a direct occurrence of money laundering by MEMEL, via a money transfer to “Futuro Inteligente Lda.”, claiming the need for “space leasing”. If before there was a lack of decisive evidence to suspect the targeted individuals and ascertain MEMEL’s role in this mysterious investigation, now it is clear there is enough substantial and tying evidence, pertaining multiple entities and actors, to demand formal interrogations and deep investigation within the mentioned companies. There is the need to either confirm or deny the contents of the e-mail, through these suggested investigations.

Finally, there is mention of physical evidence stored in a CTT Locker, in a specific address. ([See here](#)).

Given these findings, several immediate and medium-term steps are recommended to confirm the extent of collaboration and fully map the infrastructure behind the phishing operation, and to obtain further insights into MEMEL’s affairs..

- The top priority is to obtain a full forensic image of the remote host referenced as muskiboi. This server is mentioned directly in `deploy.sh` and `.bash_history`.
Investigators should attempt to recover:
 - the original zyra.zip archive,
 - web server configuration and logs (e.g., Apache/Nginx),
 - SSH/SCP authentication logs (`/var/log/auth.log`, `secure`, `.bash_history`), and

- any credential files left in /home/muskyboi/.
These artifacts will confirm when and how the phishing site was launched.

- Retrieve logs from any intermediate routers or gateways that may reveal IP addresses, timestamps, and SSH session metadata between johnnymusk and the remote host. This will enable correlation of connection times with command timestamps found in .bash_history and in the thebasement IRC log.

- The thebasement.09-23.log file contains rich contextual information: conversation timestamps, user aliases, URLs, and references to repositories. Parsing and correlating this information can help identify additional collaborators or servers. Particular attention should be given to user handles such as CatDucky, who appears in direct conversations with johnnymusk.

- All critical artifacts recovered including fst-gang.png, parking.db, wpp.png, FinancialReport.pdf, and their copies should have their SHA-256 values recalculated and confirmed. Each hash must be logged in a verified inventory to ensure legal admissibility and facilitate cross-reference with data recovered from other systems.

- Inspect the .git/config and commit metadata to determine the origin of the repository and the authors of specific commits. This step could reveal remote URLs, usernames, or commit email addresses that link to other participants or organizations.

- Submit formal requests or subpoenas to the hosting provider responsible for the muskiboi server to obtain account registration details, payment information, and complete connection logs. These details can confirm who financed or maintained the hosting infrastructure.

- Conduct structured interviews with the identified local users johnnymusk (João Musk) and ironcaesar to clarify their roles, timelines, and relationships to each other and to the remote infrastructure and the answers should be compared with forensic timelines reconstructed from command histories and file timestamps.

- Investigate the existence of the claimed “physical evidence” in the described address and seek to uncover the identity of individual “CatDucky”, potentially attempting to persuade the individual João Musk into collaborating in this investigation, perhaps using the (most likely) attempted and conducted crime of phishing as leverage in these negotiations.

- Ascertain the provenance and authenticity of the mentioned evidence, and, if applicable in light of the previous steps, engage in a formal investigation of companies MEMEL and “Futuro Inteligente Lda.”.

4 Appendices

Transcript of the conversation between users “johnnymusk” and “CatDucky”, using “irssi”:

--- Log opened Tue Sep 23 01:05:17 2025

01:05 -!- johnnymusk [~johnnymusk@freenode-uip.cj2.jjp4uo.IP] has joined #thebasement

01:05 -!- Irssi: #thebasement: Total of 2 nicks [1 ops, 0 halfops, 0 voices, 1 normal]

01:05 -!- CatDucky [~jose@freenode-uip.cj2.jjp4uo.IP] has joined #thebasement

01:05 -!- Irssi: Join to #thebasement was synced in 8 secs

01:05 < johnnymusk> Hey Cat, how've you been doing?

01:05 < CatDucky> Hii João, I'm great, Poland has been treating me well! This comp is killing me tho
ahahah

01:06 < johhnymusk> Oh right you're on a CTF comp! Must be nice to be smart.

01:06 < CatDucky> Oh don't be mean to yourself, what about all those cool tricks I taught you?

01:06 < johhnymusk> Yeah right, if it wasn't for my broken leg, I'd be there in Poland with you...

01:06 < CatDucky> What! You broke your leg??

01:07 < johhnymusk> Yeah, I was biking on one of those ZYRA bikes when all of a sudden it just started
speeding up and I couldn't hit the breaks! I tried to sue them but nothing came of it... I'll show them

01:07 < CatDucky> What are you planning johnny?

01:07 < johhnymusk> I will steal their client's credentials and get free credits for me! MUAHAHAH [2]

01:08 < CatDucky> That... is very much illegal, I know I taught you some stuff but please don't use it for
harm.

01:08 < johhnymusk> Ah, don't worry, they'll never catch me.

01:08 < CatDucky> Sure, I'll believe that. How are you planning on doing that anyways?

01:08 < johhnymusk> You know how I'm doing my thesis with professor Nuno right? They got this
machine at Syssec that I can use to host a phishing website, no one will know it was me.

01:09 < CatDucky> Again João, be very careful, that is *illegal*, I don't want nothing bad to happen to you,
and because of such a petty thing.

01:09 < johhnymusk> Don't worry, about me Cat, I'll show you how good I am! Speaking of which, what
are you planning on doing the day you arrive from Poland? *smirk*

01:09 < CatDucky> Don't even start it, MuskyBoi...

--- Log closed Tue Sep 23 01:09:53 2025

--- Log opened Tue Sep 23 01:21:44 2025

01:21 -!- johhnymusk [~johhnymus@freenode-uip.cj2.jjp4uo.IP] has joined #thebasement

01:21 -!- Irssi: #thebasement: Total of 3 nicks [1 ops, 0 halfops, 0 voices, 2 normal]

01:21 -!- Irssi: Join to #thebasement was synced in 8 secs

01:21 < johhnymusk> Yoyo Cat, Look I just got an e-mail from you... what's that about?

01:22 < CatDucky> I don't have much time right now, but I got that e-mail... and uh... I think you might be
able to need those files

01:22 < johhnymusk> I thought you said that what I was doing was illegal.

01:22 < CatDucky> And it is! Be careful

01:22 < johhnymusk> Why are you helping me out right now?

01:23 < CatDucky> Well, as I said I don't have much time to look into those myself, since I'm in the comp
right now, and I know deep down that you're doing this cause you want to make what's right!

01:23 < johhnymusk> Yeah I do, I'm glad you understand, I'm still just a little confused with all of this.

01:23 < CatDucky> We'll talk more when I get back home

01:23 < johhnymusk> Wait what?

--- Log closed Tue Sep 23 01:24:03 2025

--- Log opened Tue Sep 23 01:41:50 2025

01:41 -!- johhnymusk [~johhnymus@freenode-uip.cj2.jjp4uo.IP] has joined #thebasement

01:41 -!- Irssi: #thebasement: Total of 3 nicks [1 ops, 0 halfops, 0 voices, 2 normal]

01:41 -!- Irssi: Join to #thebasement was synced in 8 secs

01:42 < johhnymusk> Cat, I understand now, private messages among people who don't want to pay parking; Reports that takl AI calibration; A piece of code clearly stating preferential plates; And a database that confirms that some plates are not even paying parking! And a LOT of money involved! MEMEL is rigging the system in their favor! And this "Futuro Inteligente" is also behind this.

01:43 < CatDucky> What, really? That is very dangerous, we don't know who we are up against Johnny.. Don't do anything illegal please.

01:43 < johhnymusk> Don't worry Cat, I won't get caught, I just have to hide these files somehow...

01:44 < CatDucky> I heard about a good github repo for this, I think the github name was something like shadowexploit, look for FileHider or HideFiles something like this. [\[1\]](#)

01:44 < johhnymusk> RIGHT! You're the best CAT!

--- Log closed Tue Sep 23 01:44:18 2025

--- Log opened Tue Sep 23 02:04:33 2025

02:04 -!- johhnymusk [~johhnymus@freenode-uip.cj2.jjp4uo.IP] has joined #thebasement

02:04 -!- Irssi: #thebasement: Total of 3 nicks [1 ops, 0 halfops, 0 voices, 2 normal]

02:04 -!- Irssi: Join to #thebasement was synced in 8 secs

02:04 < johhnymusk> Cat Cat, All is ready!

02:04 < CatDucky> Johnny, please be very careful, you can go to jail for this...

02:05 < johhnymusk> Don't you worry, everything is secured and deployed, ZYRA is gonna suffer now. [\[3\]](#)

02:05 < CatDucky> Don't forget to delete all of the files off of your machine, use a command called SRM.

02:05 < johhnymusk> Got it! Thank u.

--- Log closed Tue Sep 23 02:05:34 2025

"deploy.sh" script, used to deploy the website

```
#!/bin/bash
```

```
set -e
```

```
echo "Zipping Zyra Website"
```

```
cd ~/Documents
```

```
zip -r ~/zyra.zip ZYRA
```

```
echo "Copying Zip To INESC machine"
```

```
scp ~/zyra.zip csf:/home/muskyboi/
```

```
echo "Initializing Zyra Website"
ssh -t csf "
    rm -rf Website/ZYRA; \
    unzip -o /home/muskyboi/zyra.zip -d Website; \
    cd Website/ZYRA; \
    npm install; \
    npm install vhost; \
    sudo node server.js
"
```

E-mail mentioning secret info (found in the directory /home/johnnymusk/.thunderbird/)

Extract of the e-mail:

Date: Tue, 23 Sep 2025 00:20:35 +0000

To: "sussyjohnnyhacker@gmail.com" <sussyjohnnyhacker@gmail.com>

From: itscatducky <itscatducky@proton.me>

Subject: Fw: You need to know this.

Message-ID:

<_pha8qoCPQgdpQyuRRUcajDNe7m4vzOJ3AjW8LZZNh2VfcLEdAJBJBOxWdJTNTvbU9LZBqP75MTFcV4BfnGuz3bp0f64RifNpol9s5s6l8=@proton.me>

In-Reply-To: <O_nu1BN--N-9@tutamail.com>

References: <O_nu1BN--N-9@tutamail.com>

Feedback-ID: 160066688:user:proton

X-Pm-Message-ID: d4baa28609f61a2a7c0fab2d17e4f4c94c96ed98

MIME-Version: 1.0

Content-Type: multipart/alternative;

Decoded slice of the e-mail:

Enviado com um e-mail seguro do [Proton Mail](https://proton.me/mail/home).

----- Forwarded Message -----

De: totallyaveragedude@tutamail.com <totallyaveragedude@tutamail.com>

Data: em terça-feira, 23 de setembro de 2025 às 01:10

Assunto: You need to know this.

Para: Itscatducky <itscatducky@proton.me>

> Listen up,

>

> I'm dropping this on you because you need to know. There's something big going on, something no one's talking about. Ever heard of MEMEL's "smart parking"? It's not smart. It's a setup.

>

> The algorithm? Rigged. Tourists bled dry, allies spared, whole neighborhoods marked "premium" just to squeeze money. And the cash? Washed clean through a ghost company called Futuro Inteligente, Lda.. Looks like AI consulting, but it's just a funnel straight into Trocos's pocket.

>

> I've got the proof. Documents, logs, invoices, the whole thing. Not online, not traceable. I stashed it for you.

>

> Here's where:

> LockerLocky CTT IST Lisbon

> Av. Rovisco Pais 1

> Locker 03

> Code: 666

>

> Go, pick it up, dig into the files. But move carefully. This isn't info they want in the open.

>

> — Your anonymous buddy

>

> --

> Protegido com Tuta Mail:

> <https://tuta.com/free-email>

Possibly stolen credentials (sample found in the backups)

Email: aaa@gmail.com, Password: ssdsdw

Email: bbbb@gmail.com, Password: 1234

Email: fdfdf@gmail.com, Password: ssss

Email: sasa@gmail.com, Password: fsdfd

Email: www@gmail.com, Password: aaa